

1 Part 4: Constructing Estimators

2 Summary

3 Recall that typically a probability model is **incompletely specified**. I.e.,
4 a particular model is assumed, but there are still unknown components,
5 often real-valued **parameters**. We use available data (a **sample**) to estimate
6 these parameters.

7 In this Part we describe methods for constructing estimators for unknown
8 parameters.

1 Terminology Review

2 The **parameters** are the constants that distinguish probability models in the
3 same “family.” Examples of “families” include “normal,” “exponential,”
4 and so forth.

5 A **statistic** is a function of the RVs being used to model the observable data.
6 Examples include the sample mean, the sample variance, etc. A statistic is
7 a **random variable**.

8 An **estimator** is a statistic which is used to estimate an unknown parame-
9 ter. It is also a **random variable**.

10 The “hat notation” is standard for denoting an estimator: $\hat{\beta}$ is an estimator
11 for β , for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its
12 estimator.

- 1 An estimator $\hat{\theta}$ is said to be **an unbiased estimator for θ** if $E(\hat{\theta}) = \theta$.
- 2 The variability in an estimator (its precision) is naturally quantified by its
- 3 variance, $V(\hat{\theta})$.
- 4 The **standard error (SE) of $\hat{\theta}$** is the square root of its variance. The SE is of-
- 5 ten reported along with an estimate in order to give a sense of the amount
- 6 of error one should anticipate in that estimate.

1 **The Method of Moments**

- 2 Simple cases such as estimating the population mean and estimating the
- 3 population proportion can be misleading: In those cases it is “obvious”
- 4 what the natural estimator is, since one can use the sample mean or sample
- 5 proportion as the estimator, respectively.
- 6 In other cases, the choice of estimator is not so clear.
- 7 We first consider a simple approach to constructing estimators, the **method**
- 8 **of moments**.

1 First, recall that for any positive integer k , the quantity $E(X^k)$ is called the
2 k^{th} moment for the distribution of X . We might call this the “ k^{th} population
3 moment” to be more clear.

4 Imagine that I have random variables $X_1, X_2, \dots, X_n$ that model the ob-
5 servable sample. Then the k^{th} sample moment is

$$6 \quad M_k = \sum_{i=1}^n X_i^k / n$$

7 Note that M_k is our best estimator for $E(X^k)$, for instance $E(M_k) = E(X^k)$

1 Now, suppose that $X_1, X_2, \dots, X_n$ are i.i.d. each with density $f_X(x; \theta)$.

2 Here, $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ is a vector of parameters.

3 Set up a system of p equations, setting the first p sample moments equal to
4 the first p population moments:

$$\begin{aligned} 5 \quad M_1 &= E(X) \\ 6 \quad M_2 &= E(X^2) \\ 7 \quad &\vdots \\ 8 \quad M_p &= E(X^p) \end{aligned}$$

10 We note that the sample moments are all quantities we can determine from
11 the available sample. We also note that the population moments are func-
12 tions of the unknown parameters.

1 We can now solve this system of equations for $\theta_1, \theta_2, \dots, \theta_p$:

$$2 \quad \theta_1 = g_1(X_1, X_2, \dots, X_n)$$

$$3 \quad \theta_2 = g_2(X_1, X_2, \dots, X_n)$$

$$4 \quad \vdots \quad \vdots$$

$$5 \quad \theta_p = g_p(X_1, X_2, \dots, X_n)$$

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7 These equations are built on the (incorrect) belief that $M_i = E(X^i)$.

8 In fact, $M_i \approx E(X^i)$, so

$$9 \quad \theta_i \approx g_i(X_1, X_2, \dots, X_n)$$

10 and we construct our estimator as

$$11 \quad \hat{\theta}_{i,\text{MOM}} = g_i(X_1, X_2, \dots, X_n)$$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. from the Exponential(λ)
 2 distribution. (Recall that this means that $E(X_i) = 1/\lambda$.) What is the method
 3 of moments estimator for λ ?

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- 1 **Exercise:** What is the result if, instead of the first moment, the second mo-
- 2 ment is used to construct the estimator for λ in the previous example?

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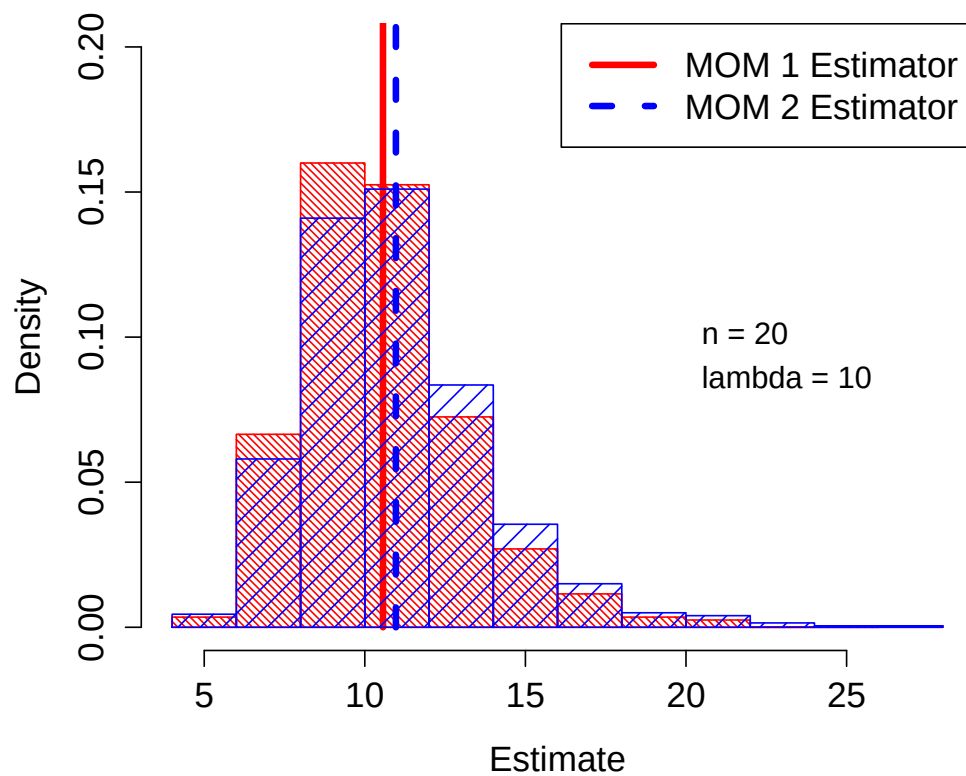
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Comparison of MOM Estimators



- 1 **Exercise:** Is the method of moments estimator found above (using the first
2 moment) an unbiased estimator for λ ? Can you adjust it to make it an
3 unbiased estimator?

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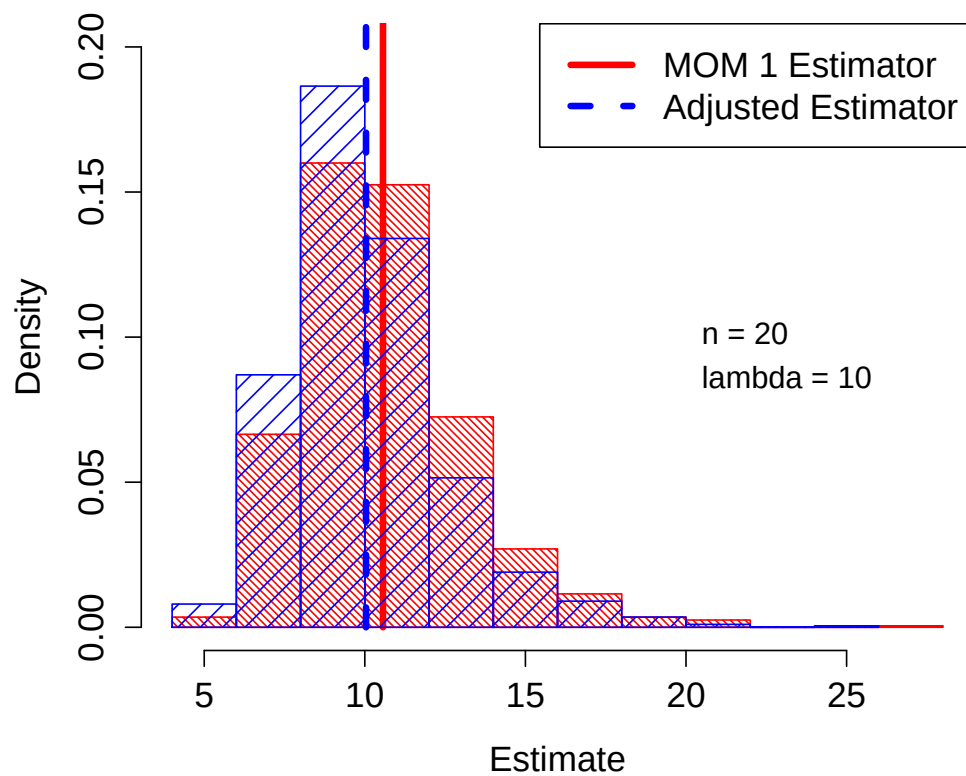
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Comparison of MOM Estimator and Adjusted MOM



- 1 The comparisons made above between the three estimators are based **en-**
- 2 **tirely on simulations.** There is **no** observed data involved at all. This is
- 3 the **frequentist approach** to assessing estimators: **how does the estimator**
- 4 **perform in repeated use?**
- 5 As seen above, simulations are a useful tool for cases where the mean and
- 6 variance derivations are not available.

- 1 **Exercise:** Interpret the output below. What is the approximate MSE of each
- 2 of the three estimators?

```
> print(c(mean(mom1), mean(mom2), mean(adju)))  
[1] 10.56553 10.96759 10.03725
```

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> print(c(var(mom1), var(mom2), var(adju)))  
[1] 6.574420 8.224773 5.933414
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1 Maximum Likelihood Estimation

2 **Maximum likelihood estimation (MLE)** is a powerful, general approach to
3 constructing estimators.

4 We will begin the explanations with simple, one-dimensional examples,
5 but the concepts extend to complex, high-dimensional models in a natural
6 manner.

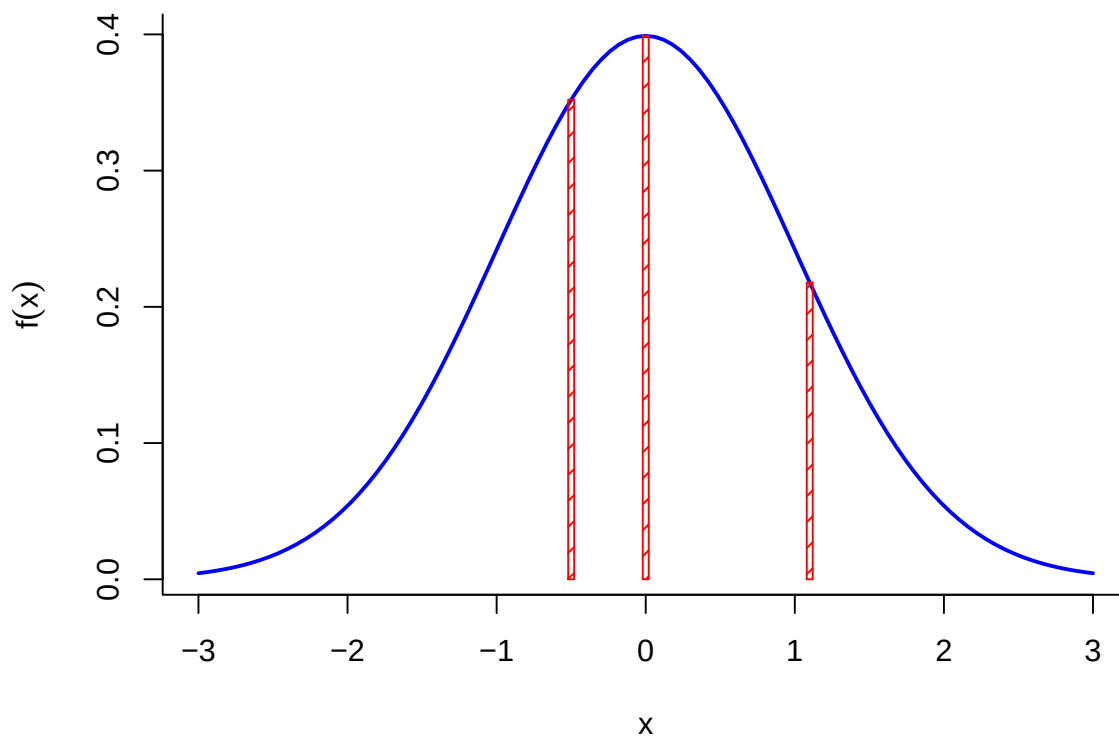
7 Importantly, there is also a general way of calculating standard errors of
8 MLEs, and, under a wide range of conditions, the MLE is approximately
9 normally distributed. This makes inference with MLEs straightforward.

1 Basic Setup

2 Recall that if X is a continuous random variable with density f_X , that does
3 **not** mean that $P(X = a) = f_X(a)$, i.e., the density does **not** return proba-
4 bilities. Indeed, $P(X = a) = 0$ for all a .

5 **It is true** that the density is higher in regions where there is the most
6 probability: For fixed, small $\epsilon > 0$, if we want to find a to maximize
7 $P(a - \epsilon \leq X \leq a + \epsilon)$, then we would look for the a that maximizes $f_X(a)$.

8 See the figure on the following slide. The standard normal density is de-
9 picted. The most likely values are close to zero, where the density is maxi-
10 mized.

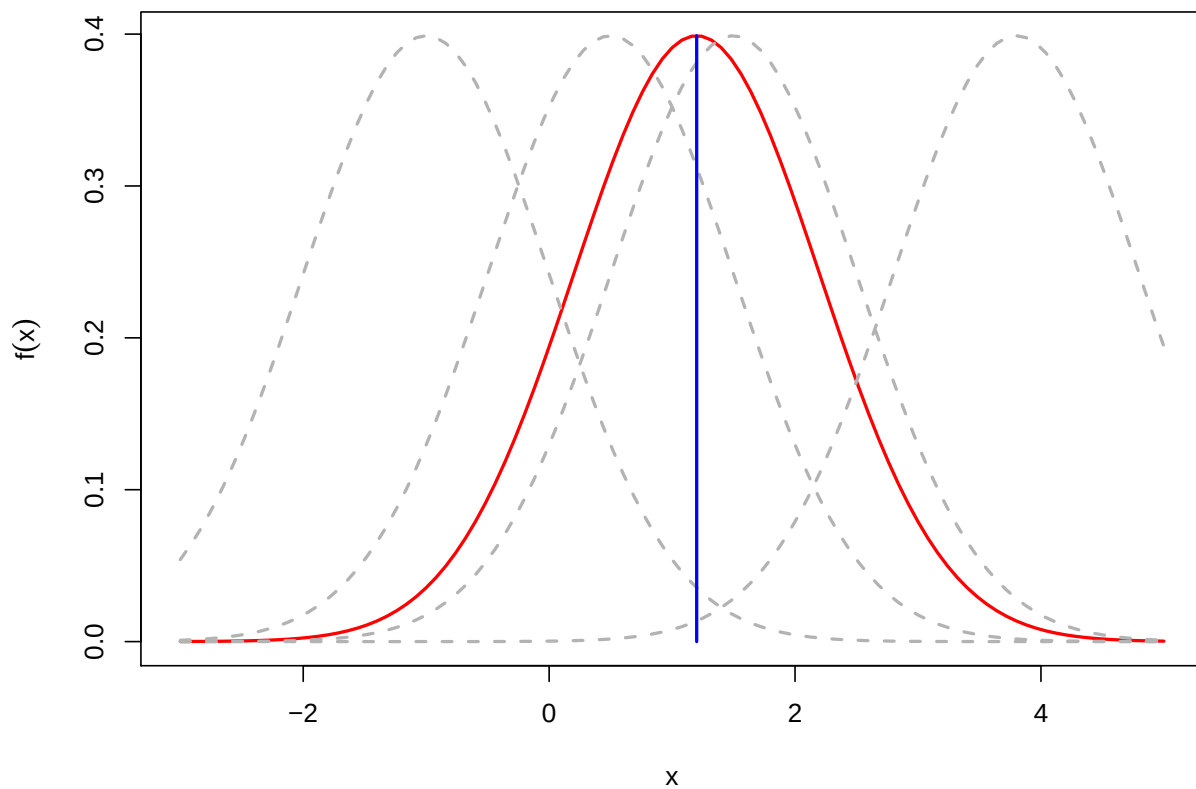


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1 As previously stated, we use probability distributions such as the normal
2 to “model” the real processes that generate data. So, if we assume that
3 our observable data is drawn from the standard normal distribution, the
4 density above specifies where we are more or less likely to see our data
5 value.

6 We can turn this idea around: Suppose that we observe data $x = 1.2$,
7 and we are willing to assume that it came from a normal distribution with
8 mean μ and variance one. What value of μ makes the observation $x = 1.2$
9 the most likely?

10 See the figure on the following slide. Different choices of μ are displayed.
11 It's not difficult to conclude that when $\mu = 1.2$, the **likelihood** is maxi-
12 mized.



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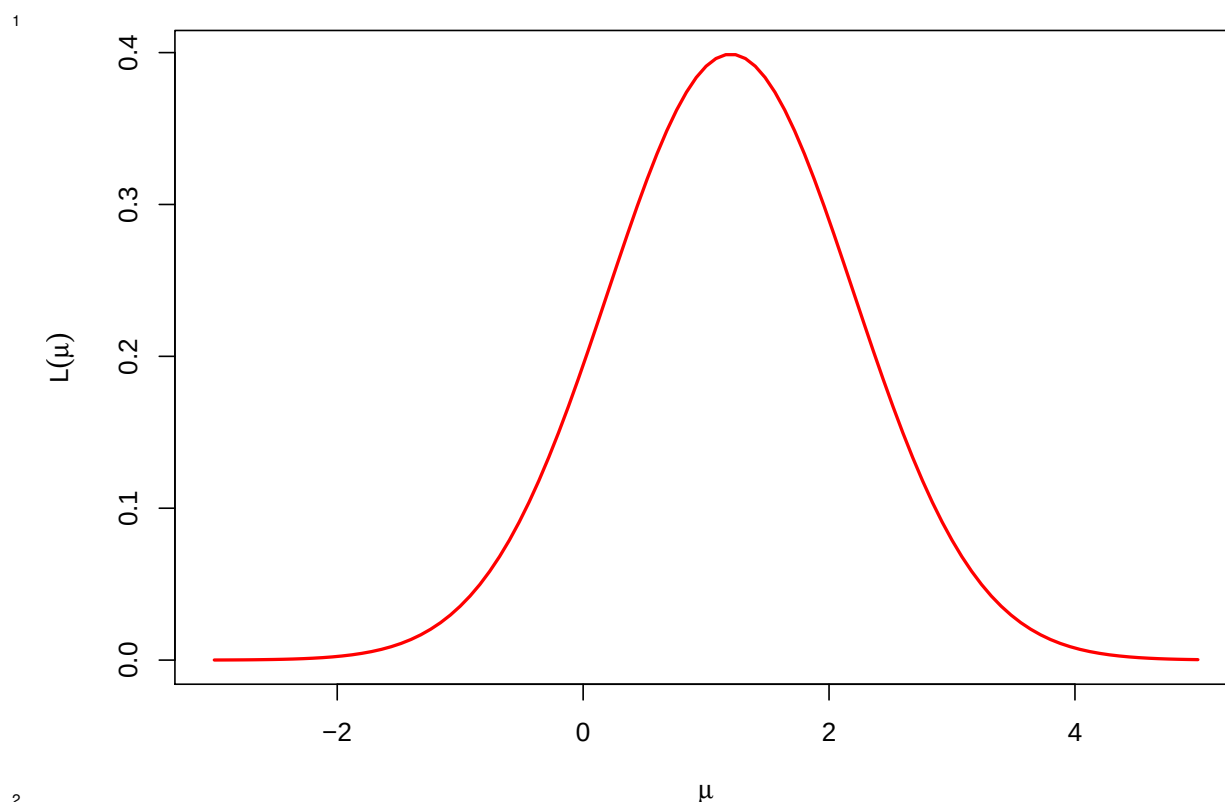
1 The **likelihood function** is constructed by calculating the distribution for
 2 the data, and then **holding the data constant** and allowing the parame-
 3 ter(s) to vary.

4 In the simple example above, we assume that we observe a single data
 5 point x that was generated from a normal distribution with mean μ and
 6 variance one. Hence, the distribution is

$$7 \quad f(x; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2}\right)$$

8 If we fix the data $x = 1.2$, and think of this as a function of μ , then we arrive
 9 at our likelihood function:

$$10 \quad L(\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1.2 - \mu)^2}{2}\right)$$



- 1 The **maximum likelihood estimator** for the parameter is the choice of the
2 parameter which maximizes the likelihood function.
- 3 Heuristically, we ask: Among all possible choices for the parameter, which
4 one makes our observed data the most likely?
- 5 It is worth stressing that the likelihood function is **not** itself a probability
6 distribution. It is **not** specifying the relative “probabilities” for different
7 values of the parameter.
- 8 Remember that in the frequentist approach, the unknown parameter is a
9 constant, and not random.

1 **Exercise:** An urn contains black and white balls. All that we know is that
2 the balls have a color ratio of 1 to 3. (We do not know if there are more
3 white or more black balls.) We draw three balls **with replacement** from
4 the urn. Let X equal the number of black draws. Derive the likelihood
5 function and the MLE as a function of X .

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1 **Exercise:** Now suppose that θ equals the proportion of balls in the urn
2 which are black. We know that $0 < \theta < 1$. We draw ten times with replace-
3 ment, and count seven black balls and three white balls. Derive the MLE
4 for θ .

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1 The preceding examples involved a single, observable random variable. In
2 most cases, we have a sequence of random variables in the model.

3 A common case is where $X_1, X_2, \dots, X_n$ are i.i.d. with density $f_X(x; \theta)$,
4 where θ is the parameter to be estimated. In this case, the joint distribution
5 of the random variables is

$$\prod_{i=1}^n f_{X_i}(x_i; \theta)$$

7 and the likelihood function is

$$L(\theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta)$$

1 Finding the MLE Analytically

2 In some situations the MLE can be derived analytically, usually via calcu-
3 lus, but sometimes via direct inspection of the likelihood.

4 One “trick” that is often very helpful: The likelihood function is positive
5 at its maximum. Hence, the likelihood is maximized at the same θ as does
6 the logarithm of the likelihood.

7 Hence, we often work with the **log likelihood** instead of the likelihood.
8 This is often denoted $\ell(\theta) = \log L(\theta)$.

9 In the case where $X_1, X_2, \dots, X_n$ are i.i.d. $f_X(x; \theta)$, then

$$\ell(\theta) = \log \prod_{i=1}^n f_{X_i}(x_i; \theta) = \sum_{i=1}^n \log f_{X_i}(x_i; \theta)$$

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Exponential(λ)
2 distribution. Derive the MLE for λ .

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- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Beta($\alpha, 1$) distri-
2 bution. Derive the MLE for α . Compare this with the MOM estimator.

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- 1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. with the Normal(μ, σ^2) dis-
2 tribution. Derive the MLE for μ and σ^2 .

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- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Uniform}(0, \theta)$. Derive the
2 MLE for θ . What is the MOM in this case?

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